1. **Aim of the practical:** Write a program based on logistic regression and make the sigmoid graph using sigmoid function in JupyterNotebook.

2. **Objective:** To implement and apply a regression algorithm on a given dataset in order to study the relationship between independent features and the target variable, and to evaluate the model performance using suitable metrics (like Mean Squared Error and R^2 score).

3. **Algorithm/Flowchart:**

```
import pandas as pd
import seaborn as sns
df=pd.read_csv('Social_Network_Ads.csv')
print(df)
```

```
df["Purchased"]=df["Purchased"].astype('category')
df["Purchased"]=df["Purchased"].cat.codes
print(df)
```

```
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
plt.scatter(df['Age'],df['Purchased'])
sns.regplot(x=df['Age'],y=df['Purchased'],data=df,logistic=True,ci=None)
x=df[["Age"]].values.reshape(-1,1)
y=df[["Purchased"]]
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.1)
from sklearn.linear_model import LogisticRegression
reg=LogisticRegression()
reg.fit(x_train,y_train)
reg.predict(x_test)
y_test=x_test*reg.coef_+reg.intercept_
print(y_test)
```

```python
from scipy.special import expit
sigmoid=expit(y_test)
print(sigmoid)
```

4. Result/Output:

```
      User ID  Gender  Age  EstimatedSalary  Purchased
0    15624510    Male   19            19000          0
1    15810944    Male   35            20000          0
2    15668575  Female   26            43000          0
3    15603246  Female   27            57000          0
4    15804002    Male   19            76000          0
..        ...     ...  ...              ...        ...
395  15691863  Female   46            41000          1
396  15706071    Male   51            23000          1
397  15654296  Female   50            20000          1
398  15755018    Male   36            33000          0
399  15594041  Female   49            36000          1

[400 rows x 5 columns]
```

```
      User ID  Gender  Age  EstimatedSalary  Purchased
0    15624510    Male   19            19000          0
1    15810944    Male   35            20000          0
2    15668575  Female   26            43000          0
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..        ...     ...  ...              ...        ...
395  15691863  Female   46            41000          1
396  15706071    Male   51            23000          1
397  15654296  Female   50            20000          1
398  15755018    Male   36            33000          0
399  15594041  Female   49            36000          1

[400 rows x 5 columns]
```
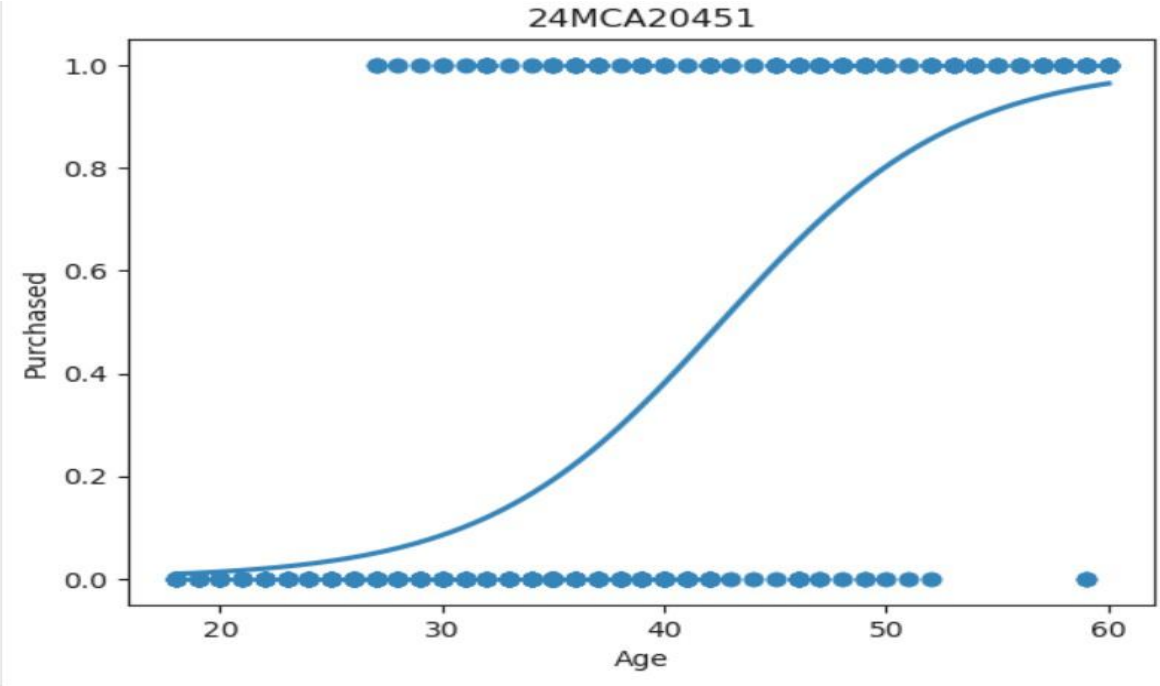

```
[[0.33544287]
 [0.93394846]
 [0.96220685]
 [0.90524481]
 [0.15924427]
 [0.66565554]
 [0.47612976]
 [0.03142541]
 [0.02145129]
 [0.21895188]
 [0.29323835]
 [0.0259764 ]
 [0.13471466]
 [0.90524481]
 [0.18727368]
 [0.78189027]
 [0.86585913]
 [0.8870414 ]
 [0.02145129]
 [0.66565554]
 [0.8870414 ]
 [0.03797289]
 [0.95439512]
 [0.42761413]
 [0.02145129]
 [0.03797289]
 [0.25431184]
 [0.33544287]
 [0.03142541]
 [0.01202819]
 [0.25431184]
 [0.15924427]
```



5. Learning outcomes (What I have learnt):

- Understood the concept of regression and how it helps in predicting continuous values.
- Learned to load and preprocess a dataset for regression tasks.
- Implemented **Linear Regression** using scikit-learn in Python.
- Evaluated model performance using metrics like **Mean Squared Error (MSE)** and **R^2 score**.